

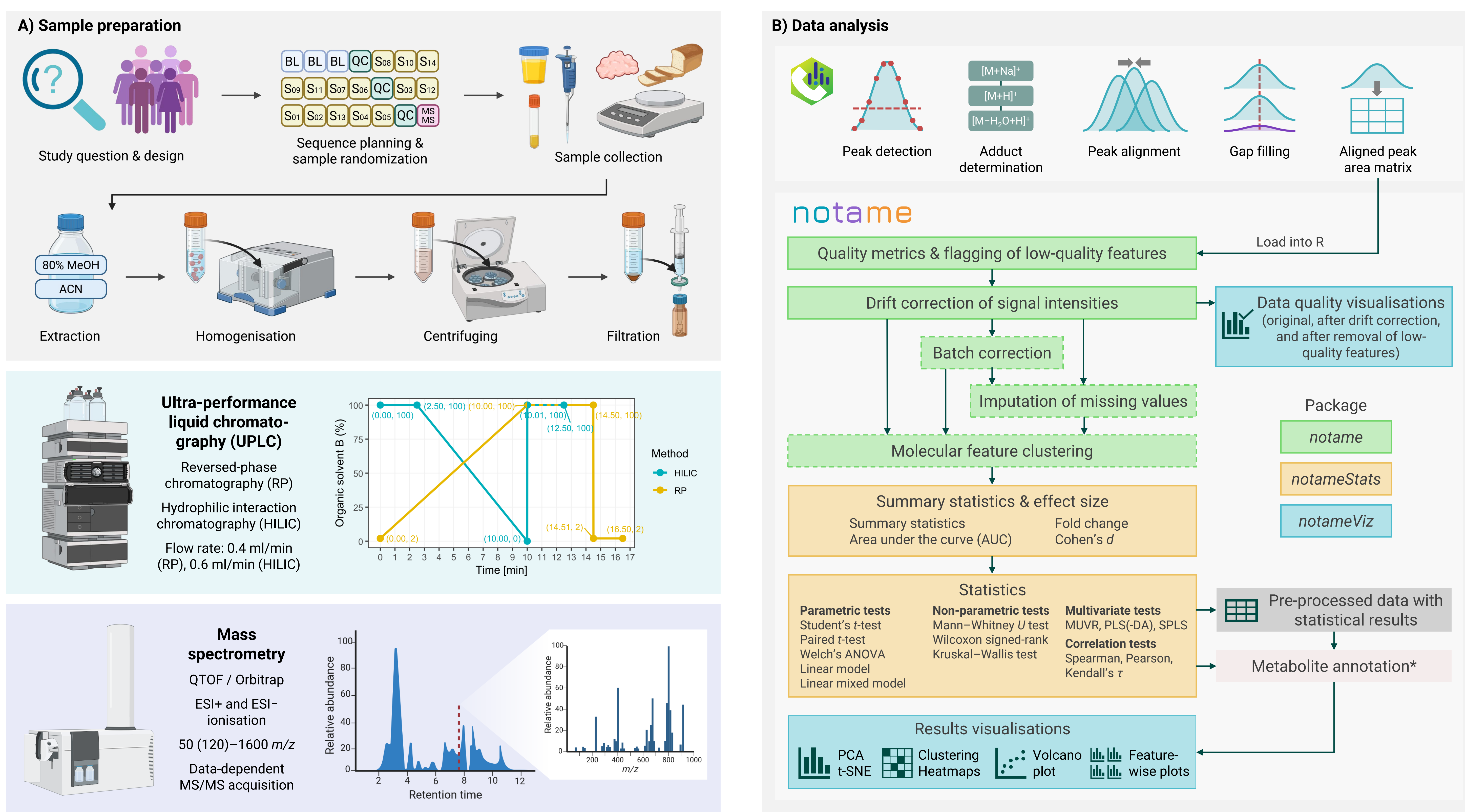
notame – a complete non-targeted LC-MS metabolomics workflow

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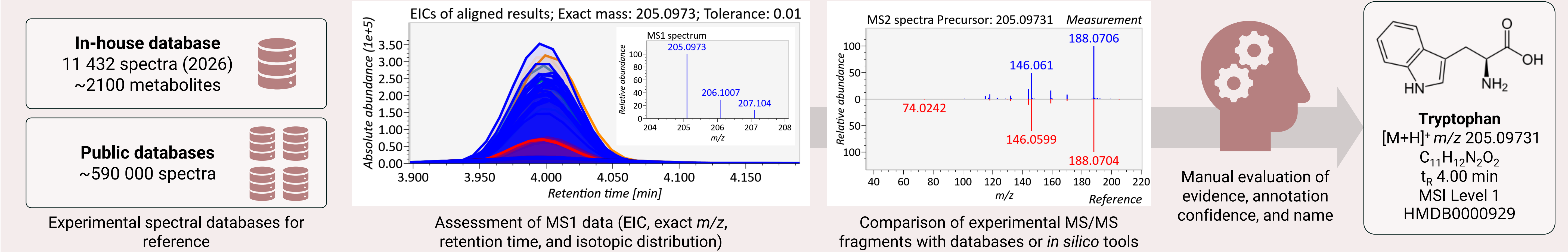
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Introduction

- Metabolomics aims to comprehensively measure all the small molecules in a biological system to determine its biochemical state
- Uniform sample preparation, wide analytical coverage, quality-controlled data pre-processing, and transparent results communication are key requirements for an impactful metabolomics study
- notame* is a peer-reviewed workflow [1] and R package bundle [2] that aims to cover all aspects of a metabolomics study from study design and sample preparation to feature selection and visualizations
- It is based on the LC-MS metabolomics method developed at the University of Eastern Finland in the early 2010s [3] and the R packages developed 2017 onwards together with UEF and Afekta Technologies Ltd, used in nearly 100 publications



*Metabolite annotation



References

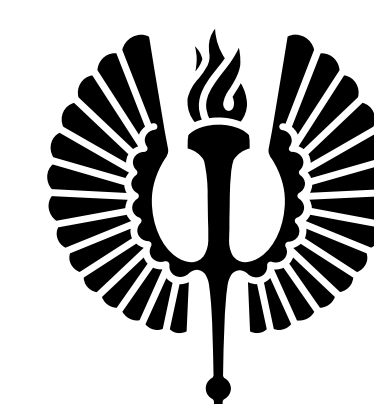
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- [2] Koistinen et al. *Comprehensive LC-MS metabolomics data processing with notame R/Bioconductor package*. Preprint 2026. DOI: 10.5281/zenodo.18621243
- [3] Hanhineva et al. *Nontargeted Metabolite Profiling Discriminates Diet-Specific Biomarkers for Consumption of Whole Grains, Fatty Fish, and Bilberries in a Randomized Controlled Trial*. *J. Nutr.* 2015

Acknowledgements

Anton Klåvus
Jussi Paananen
Seppo Auriola, UEF
Marko Lehtonen, UEF
David Broadhurst, Edith Cowan University, Australia



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